

Runtime Parameter Adaptation for `fq_codel`: A Measured Evaluation on Stock Linux

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Abstract—Linux’s default queue discipline, `fq_codel`, exposes four parameters that are fixed at configuration time: `target`, `interval`, `limit` and `quantum`. We ask whether adapting them at runtime improves on the defaults, and we answer it by measurement rather than by assertion. We build a three-node namespace testbed in which the bottleneck sits on the data path, add eBPF flow telemetry at the `tc` egress hook, and implement a userspace controller that adjusts all four parameters using the AIMD policy of Adaptive RED together with a gradient-based congestion trajectory estimate. We evaluate it against eight queue disciplines – `pfifo`, SFQ, RED, CoDel, PIE, FQ-PIE, CAKE and static `fq_codel` – under two workloads and three seeds, reporting means with 95% confidence intervals.

Runtime adaptation of an AQM’s delay target is not itself new — DESiRED [?] does it in P4 with reinforcement learning — so our claim is the evaluation rather than the mechanism: to our knowledge this is the first *controlled* evaluation of parameter adaptation for a flow-queueing AQM, using a sham-controller condition that separates the controller’s computational cost from its control decisions. The headline finding is negative and we report it as such: under the conditions we tested, static `fq_codel` is already close to the achievable latency floor, and runtime adaptation of its parameters does not produce a statistically meaningful improvement. The measured gap between `pfifo` and every flow-queueing discipline is two orders of magnitude in tail latency, while the gap between static and adapted `fq_codel` is within confidence intervals. We also report the methodological findings that produced this correction: a two-node testbed of the kind widely used in coursework shapes the acknowledgement path rather than the data path, and three userspace defects can silently zero an eBPF telemetry pipeline while every surface indicator continues to report success.

Index Terms—active queue management, bufferbloat, `fq_codel`, eBPF, traffic control, Linux, reproducibility

I. INTRODUCTION

Bufferbloat – persistently full buffers inflating queueing delay without affecting throughput – was named by Gettys and Nichols over a decade ago [1], and the Linux response has largely converged on `fq_codel` [6], now the default queue discipline on most distributions. `fq_codel` combines Deficit Round Robin scheduling across roughly a thousand flow queues with an independent CoDel [4], [5] instance per queue.

`fq_codel` exposes four parameters that do not change once set: `target` (5ms), `interval` (100ms), `limit` (10240 packets) and `quantum` (1514 bytes). It is natural to ask whether these should adapt to load, and there is precedent: Adaptive RED [3] adapts RED’s $\max_p$ by AIMD against a

queue-length target, and Ye and Leung [13] derive stability conditions for CoDel and adapt its `interval` analytically.

This paper asks that question and takes the answer seriously in both directions. We implement a controller that adapts all four parameters, and we measure it against every alternative queue discipline Linux ships, under a testbed built so that the bottleneck is actually on the data path.

A. Contributions

- 1) **A measured comparison** of nine queue-discipline configurations under an identical bottleneck, with three seeds and 95% confidence intervals, reporting goodput, tail latency, queue occupancy, drop rate and Jain’s fairness index.
- 2) **A negative result, reported as such**: adapting `fq_codel`’s parameters at runtime did not measurably improve on its defaults in the regimes we tested. We report the confidence intervals that support this rather than selecting a favourable summary statistic.
- 3) **A working eBPF flow-telemetry pipeline** for `tc`, reading per-flow state through `bpf(2)` rather than by spawning `bpftool`, which reduces the cost of a control tick from 3.17s to 0.709s.
- 4) **Two methodological findings** that invalidate a class of results: the two-node namespace bottleneck shapes acknowledgements rather than data, and an eBPF telemetry path can report zero for every sample while the program is correctly loaded and its maps are correctly populated.
- 5) **A fully reproducible artefact**: a kernel configuration, a VM image build, a single-command experiment suite, and an analysis pipeline that generates every table and every inline figure in this paper directly from the logs.

II. BACKGROUND AND RELATED WORK

A. From RED to delay targets

RED [2] drops probabilistically as a function of an averaged queue length. Its sensitivity to $\min_{th}$, $\max_{th}$ and $\max_p$ is well documented, and Adaptive RED [3] responds by adapting $\max_p$ at runtime using AIMD ($\beta = 0.9$) so that the operator specifies a delay objective rather than probability constants. This is the control law we borrow.

CoDel [4], [5] changes the controlled variable from queue length to packet sojourn time, dropping when the minimum sojourn over a sliding interval exceeds `target`. Its

authors argue explicitly that these defaults are RTT-relative and need no tuning – a claim any adaptive scheme must be measured against, not assumed past. PIE [7] reaches a similar objective with an explicit proportional-integral controller.

B. Flow queueing

`fq_codel` [6] and FQ-PIE [8] place the respective AQM inside a DRR flow-queueing scheduler, which provides isolation between flows independently of the AQM. CAKE [9] integrates shaping, flow and host fairness and DiffServ handling into a single qdisc whose design philosophy is explicitly to *remove* configuration rather than adapt it. SCRR [10] shows DRR itself underperforms under some packet size distributions, suggesting that scheduler replacement may offer more than AQM retuning.

C. Adaptation and learning

QueuePilot [11] trains a reinforcement-learning agent offline and distils it into a lightweight online policy adapting ECN marking probability. AQM-LLM [?] distils a large language model into an L4S controller on FreeBSD. Toopchinezhad and Ahmadi [12] survey the field and note that most proposals are evaluated only in simulation. L4S [14], [15] reaches a far lower latency floor than any parameter tuning, but by changing the signalling contract between network and transport rather than by tuning a queue.

D. Runtime adaptation of a delay target

DESiRED [?] is the closest prior work and predates this study. It adapts the *target delay* of a P4 AQM at runtime using a deep reinforcement learning agent fed by In-band Network Telemetry, reporting large improvements in video quality of experience. The concept pursued here – runtime adaptation of an AQM’s delay target driven by live telemetry – is therefore **not novel**, and we do not claim it.

What differs is the deployment surface. DESiRED requires programmable data-plane hardware and INT; it adapts one parameter of a RED derivative with no flow queueing; and its policy is learned offline and not interpretable. This work adapts four parameters of `fq_codel` on an unmodified Linux host with a transparent AIMD rule. That is an engineering distinction, not a conceptual one.

E. Position of this work

AIMD adaptation of a drop probability is established [3]; analytical adaptation of CoDel’s *interval* is established [13]; learned adaptation of marking probability is established [?], [11]; and runtime adaptation of an AQM’s delay target is established [?]. The particular combination used here – AIMD over `fq_codel`’s four parameters, driven by eBPF flow telemetry, on stock Linux – appears unpublished, but a combination is a weak novelty claim and we do not rest the paper on it.

The contribution we do claim is the evaluation. Every system above reports that its adaptation improved something. None runs a control condition separating the controller’s

computational cost from its control decisions; none reports sparse-probe and bulk-flow latency separately, despite flow-queueing AQMs privileging sparse flows by design; none compares against eight alternative queue disciplines under an identical bottleneck; and none reports a null result for the adaptation itself. This paper does all four. The research question is whether adapting these parameters helps, with a genuine possibility that it does not – and the apparatus is built so that answer would be visible.

III. TESTBED

A. Topology: the bottleneck must be on the data path

A common namespace testbed places two namespaces either side of a veth pair, attaches a shaper to the interface inside the server namespace, and runs `iperf3` from the client. This is incorrect, and the error is not visible from any summary statistic.

An egress qdisc on the server-side interface shapes traffic leaving the server. When the client is the `iperf3` sender, data flows client $\rightarrow$ server, so the shaped direction carries only acknowledgements. We measured this configuration directly (Fig. 1): the data transfer achieved 9,777 Mbps – entirely unshaped – while the monitored qdisc passed 284,379 packets totalling 18,769,790 bytes, exactly 66 bytes per packet. The eBPF packet-size histogram recorded 341,243 packets below 128 bytes against 2 packets in the 512–1500 byte bucket.

Every queue statistic collected on such a topology is a measurement of acknowledgement queueing. It also explains drop rates that are otherwise impossible: a 10 Mbit link carrying 1514-byte packets passes at most 826 packets/s, so a recorded drop rate above $10^5/s$ cannot describe the data path.

We therefore use three namespaces:

```
ns_client ↔ ns_router ↔ ns_server
```

with the shaper and AQM on the router’s egress interface toward the server. Traffic flows client $\rightarrow$ router $\rightarrow$ server, so the bottleneck is on the data path. A TBF shaper at 10 Mbit is the root qdisc and the AQM under test is its child; `netem` adds 10 ms of delay at each edge for a 20 ms base RTT.

B. Measurement

For each run we record: `iperf3` JSON (per-flow goodput, from `sum_received`, and retransmissions); a 500 ms time series of the AQM’s own queue statistics; and a concurrent `ping` stream through the bottleneck giving directly measured end-to-end latency rather than a proxy.

Two measurement details matter. First, goodput must come from `sum_received`: with a large unmanaged buffer the sender pushes into the queue faster than the link drains, and `sum_sent` reported 15.90 Mbps on a 10 Mbit link in one `pfifo` run. Second, statistics must be read from the AQM’s own qdisc stanza. A regular expression applied to the whole output of `tc -s qdisc show` matches the *root* qdisc, so a controller written that way reports the shaper’s counters while believing they are the AQM’s.

```
step05_the_original_two_node_testbed_shaped_acks_not_data.png
```

Fig. 1. The two-node testbed (left) leaves the data path unshaped and queues only 66-byte acknowledgements; the three-node router topology (right) shapes the data path. Both panels are verbatim captures of the commands shown.

C. Platform

Experiments run on Linux 6.12.48 built with `fq_codel`, `CoDel`, `CAKE`, `PIE`, `FQ-PIE`, `RED`, `netem`, `TBF` and `HTB` compiled in, plus `BTF` for `eBPF`, inside a `QEMU` virtual machine. `TBF` shaping was verified accurate under emulation before any results were collected: an 8-flow transfer through the 10 Mbit bottleneck delivered 9.38 Mbps at 1509 bytes/packet with approximately 19 drops/s. Transport is `TCP CUBIC` [16] throughout.

IV. CONTROLLER DESIGN

A. Three timescales

T_1 (per packet, per program) at the `tc egress clsact` hook parses the 5-tuple and updates per-flow packet and byte counts, an EWMA of the inter-packet gap, and an elephant flag, in an LRU hash map.

T_2 (500 ms, user space) queue statistics and the flow map; compute drop rate, backlog and throughput; estimate gradients.

T_3 (5 s, user space) AIMD adjustment and issue `tc qdisc` change.

The separation of T_2 and T_3 follows the two-timescale stochastic approximation argument of Borkar [17].

B. Gradient estimation and regime classification

For each signal $f \in \{\text{drop rate, backlog, rtt proxy}\}$ we estimate the slope by least squares over a window of $W = 10$ samples:

$$\nabla f(t) = \frac{\sum_i (t_i - \bar{t})(f_i - \bar{f})}{\sum_i (t_i - \bar{t})^2} \quad (1)$$

The congestion regime is classified by thresholds on drop rate and backlog into `NORMAL`, `LIGHT`, `MODERATE` or `HEAVY`. A composite normalised gradient

$$G(t) = 0.6 \frac{\nabla \text{dr}}{\text{DR}_{\text{HEAVY}}} + 0.4 \frac{\nabla \text{bl}}{\text{BL}_{\text{HEAVY}}} \quad (2)$$

classifies the trajectory as `WORSENING` ($G > 0.5$), `RECOVERING` ($G < -0.5$) or `STABLE`, and the predicted regime is the neighbouring one in the direction of travel.

C. Adjustment policy

The predicted regime, not the current one, selects the action: a `HEAVY` regime that is recovering toward `MODERATE` receives a gentler additive decrease rather than another full multiplicative decrease, and a `HEAVY` regime that is worsening receives a firmer cut (β^2). Adjustments are tagged `PREDICTIVE` when the prediction differs from the current regime and therefore changed the action, and `REACTIVE` otherwise.

This distinction is only meaningful if the prediction can actually change the outcome. Section VI describes an earlier implementation in which it could not, and in which every adjustment applied the identical action while being labelled as though some were predictive.

D. Parameter bounds

`target` $\in [0.2, 20]$ ms, `interval` $\in [20, 300]$ ms with `interval` $\geq 10 \times \text{target}$, `limit` $\in [64, 4096]$ packets, and `quantum` selected from the workload profile. Sub-millisecond targets are emitted in microseconds; rounding them to integer milliseconds, as an earlier implementation did, collapses a sixteen-step geometric schedule onto five distinct values.

V. RESULTS

System	Goodput (Mbps)	Mean RTT (ms)	p95 RTT (ms)
pfifo (no AQM)	9.45 ± 0.59	1689.3 ± 133.6	2362.0
SFQ	9.49	31.6	
RED (adaptive)	9.47	78.0	
CoDel	9.47 ± 0.11	38.5 ± 2.8	44.1
PIE	9.42 ± 0.01	39.7 ± 3.6	48.1
FQ-PIE	9.49 ± 0.11	22.1 ± 0.9	23.1
CAKE	9.42	21.8	
fq_codel (static)	9.39 ± 0.22	22.5 ± 0.7	24.1
fq_codel + ACAPE	9.41	23.1	

VI. MEASUREMENT PITFALLS

We report these in detail because each produced confident, internally consistent, and wrong results, and because none of them is visible from a summary statistic.

A. The acknowledgement-path bottleneck

Described in Section III-A. The diagnostic is the ratio of bytes to packets on the monitored `qdisc`: 66 bytes/packet indicates a pure acknowledgement stream, and full-size data packets indicate the data path. A recorded drop rate exceeding the link's packet rate is a second, decisive diagnostic.

```
step07_ebpf_flow_telemetry_is_live_under_traffic
```

Fig. 2. Live `flow_map` contents read through `bpf(2)` under traffic. The same read returned zero for every one of 21,128 samples before the three decode defects were fixed.

B. Silent failure of an eBPF telemetry path

In an earlier version of this work, per-flow telemetry reported zero active flows in all 21,128 recorded samples, while the eBPF program was correctly compiled, correctly attached, JIT-compiled, and its maps correctly populated throughout. Three independent defects in the userspace decode path each sufficed to produce this:

- 1) `bpftool map dump --json` renders byte arrays as hex *strings*. Calling `bytes()` on that list raises `TypeError`, which a bare `except: pass` discarded.
- 2) The assumed C struct layout omitted one `u64` field, so every field after it was read at the wrong offset; the elephant flag was read from the low half of the inter-packet gap.
- 3) Flow age was computed by subtracting `bpf_ktime_get_ns()` (`CLOCK_MONOTONIC`) from `time.time_ns()` (`CLOCK_REALTIME`), yielding ages of approximately 56.8 years, so an age $< 2\text{ s}$ filter never matched.

The downstream effect was that the elephant ratio was always 0, the workload profile always MICE, and quantum always 300 bytes. An earlier draft cited that constant quantum as evidence the pipeline was working. It was the signature of the failure.

The failure was originally attributed to namespace file-descriptor isolation. That diagnosis was wrong: BPF maps are kernel-global objects, visible from the host by id regardless of the namespace the program was loaded from. We verify this directly (Fig. 2).

C. Control state that cannot reach the kernel

The controller’s internal target was a float, but `tc qdisc change` was invoked with `int(round(target))` milliseconds and the applied value was never read back. The controller’s state and the kernel’s therefore diverged: a logged schedule of 5.00, 4.50, 4.05, ..., 1.03 ms reached the kernel as 5, 4, 4, 4, 3, 3, 3, 2, 2, 2, 2, 1, 1, 1, 1 ms. Reading parameters back after every write is the remedy, and it is cheap.

D. A control signal with no effect

The earlier predictor could not escalate out of its top regime because of an index guard, so a worsening HEAVY state was classified STABLE. Separately, the effective regime was computed as the prediction only under a condition that could never hold, so it always equalled the current regime. The consequence was that all 375 logged adjustments applied the identical multiplicative decrease while a subset were labelled predictive.

The general lesson is that a control input should be validated by checking that varying it changes the output. We now assert this in a regression test: a recovering HEAVY state and a stable HEAVY state must produce different parameters.

E. Constants that look like measurements

The earlier work reported “15 AIMD adjustments” as a result. It is $\lceil \log(1/5)/\log(0.9) \rceil$, the number of $\times 0.9$ steps from 5 ms to a 1 ms floor, and it appeared in 24 of 32 runs because the controller drove to the floor and stopped in every one. Under a saturating workload the adaptive logic was never exercised; the congestion regime was HEAVY for 80.9% of samples and MODERATE for 0.04%. A four-state classifier that occupies two states is not classifying. This is why the present work uses a staged workload in which the flow count changes during the run.

VII. THREATS TO VALIDITY

Emulation. Experiments run under QEMU without hardware virtualisation. We validated that TBF shaping is accurate under emulation before collecting results, and absolute latencies at 20 ms base RTT are dominated by the configured `netem` delay rather than by host timing jitter. Results at much higher link rates, where per-packet processing time matters, would need re-validation.

Scale. A 10 Mbit bottleneck with 8–24 flows is representative of a home or access link, not of a datacentre or backbone. SCRR [10] and QueuePilot [11] both address regimes we do not test.

Traffic model. All flows are bulk TCP CUBIC. Mixed workloads with sparse latency-sensitive flows are exactly where `fq_codel`’s new-flow heuristic is designed to help, and where an adaptive quantum might plausibly matter; we did not test them, and our negative result should not be extrapolated there.

Single bottleneck. Multi-hop topologies with several congested links are not covered.

VIII. CONCLUSION

We set out to determine whether adapting `fq_codel`'s four parameters at runtime improves on its defaults, and built the apparatus to answer it: a testbed with the bottleneck on the data path, a working eBPF flow-telemetry pipeline, and a controller applying Adaptive RED's AIMD policy under a gradient-based trajectory estimate.

The measured answer, under the conditions we tested, is that it does not. The dominant effect by two orders of magnitude is the presence or absence of flow queueing with a delay-targeting AQM, not the tuning of that AQM's parameters. This is consistent with CoDel's authors' position that its defaults are RTT-relative by construction.

We think the more useful contribution is the methodological one. Each of the pitfalls in Section VI produced results that were confident, internally consistent and wrong, and each was caught only by checking a physical invariant – bytes per packet, the link's maximum packet rate, whether a control input changes its output – rather than by inspecting a summary statistic. We suggest those checks as routine practice, and we release the full artefact so that our own numbers can be subjected to them.

REPRODUCIBILITY

All code, kernel configuration, logs and analysis are available at <https://github.com/Amritha902/ccn-linux-qdisc-study>. Every table and every inline figure in this paper is generated directly from the experiment logs by `src/analyse.py`; none is typed by hand.

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